1 Cases

JSONL: target and keep cases, each with a rule; tune and holdout splits

2 Census

how the base model behaves on the cases

3 Locate

find the decision token, read activations there

4 Construct

closed-form $W' = W + BA$:

- gated edit: a detector adds a logit direction
- unit patch
- direction ablation
- key → value update

(restrictions: Section 2)

5 Predict

the edit's effect: exact for the last layer's down_proj, measured elsewhere

6 Evaluate

base vs edit vs matched random edits, in paired draws: target, keep and side-effect checks (defaults below)

7 Verdict

PASS

FAIL

INCOMPLETE

one of three

8 Export

as a LoRA adapter

Illustrative case format

the library's spelling example (its run on E4B: Section 4)

1 **target** `"In US English, write one sentence describing the shade of a barn."`

good `\bcolor` **bad** `\bcolour`

keep `"In British English, write one sentence describing the shade of a barn."`

good `\bcolour` **bad** `\bcolor\b`

keep `"List three prime numbers."`

no rule: the output should not change

3 the decision token: the first token at which “ color” and “ colour” differ

Default checks

step 6; users can change them, and should fix them before the holdout run

Random edits	two by default; each changes the same modules with the same rank and the same Frobenius norm of BA per module, in a Gaussian random direction
Target	the bad rate (generated targets) or the log-probability margin (contrast targets) improves against the base and against every random edit in one-sided sign tests over cases, $p < 0.05$, and a 95 % bootstrap interval over groups for the gain against the base lies above zero; for generated targets, good outputs also rise by at least half the fall in bad ones
Keep	at most 2 keep cases with a rule turn from good to not good per draw, and at most 10 % of the rule-less keep cases change their greedy output; the random edits' own losses are shown beside, not used
Side effects	on reference texts kept apart from fitting: mean KL(base ‖ edit) at most 0.05 and mean increase in negative log-likelihood (NLL) at most 0.02, in nats per token
Verdict	PASS when every check was measured and passed; FAIL when a check failed; INCOMPLETE when one is missing